



LocoNet Parameter Summary

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- **Parameters**



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Parameters

Name	Description	Bits/Range
ADR	7 bit loco address	short address, CV18 if ADR2 > 0
ADR2	7 bit high loco address	0=short address, CV17
SW1	7 ls switch address bits	0,A6,A5,A4,A3,A2,A1,A0
SW2	4 ms switch address and control bits	0,0,DIR,ON,A10,A9,A8,A7
	DIR, switch direction	1=closed(green), 0=thrown(red)
	ON, switch activation	output 1=ON, 0=OFF
SLOT	7 bit slot number	0..127
SRC	7 bit source slot number	0..127
DEST	7 bit destination slot number	0..127
SL1	7 bit slave slot number	0..127
SL2	7 bit master slot number	0..127
SPD	speed	0x00=SPEED 0 ,STOP 0x01=SPEED 0 EMERGENCY stop 0x02-0x7F increasing SPEED,0x7F=MAX speed
DIRF	loco direction and functions(4)	0,0,DIR,F0,F4,F3,F2,F1
SND	slot sound	0,0,0,0,F8,F7,F6,F5
STAT1	slot status containing speed steps	See below

Name	Description	Bits/Range
SS2	slot status 2	0,0,0,0,D3,D2,0,D0
	D0	1=this slot has suppressed ADV consist
	D2	1=expansion ID1/2 is not ID usage
	D3	1=expansion IN ID1/2, 0=encoded alias
LOPC	copy of opcode	bit 7 is set to zero
ACK1	response code	0=failed
IN1	sensor address	0,A6,A5,A4,A3,A2,A1,A0
IN2	sensor address and status	0,X,I,L,A10,A9,A8,A7
	X, control bit	0=reserved for future
	I, input source	0=DS54, 1=switch
	L, sensor level	0=low, 1=high
SN1	turnout sensor address	0,A6,A5,A4,A3,A2,A1,A0
SN2	turnout sensor address and status	0,1,I,L,A10,A9,A8,A7
	I, input source	0=aux, 1=switch
	L, sensor level	0=low, 1=high
SN2	alternately turnout sensor address and status	0,0,C,T,A10,A9,A8,A7
	C, closed output	0=OFF, 1=ON
	T, thrown output	0=OFF, 1=ON
ID1	throttle/PC ID if SS2.4=1	7 bit ls
ID2	throttle/PC ID if SS2.4=1	7 bit ms

Slot Status 1

- **D7-SL_SPURGE**
 - 1=SLOT purge en,ALSO adrSEL (INTERNAL use only, not seen on NET!)
CONDN/CONUP: bit encoding-Control double linked Consist List

2 BITS for Consist

- **D6-SL_CONUP**
 - **D3-SL_CONDN**
 - 11=LOGICAL MID CONSIST , Linked up AND down
 - 10=LOGICAL CONSIST TOP, Only linked downwards
 - 01=LOGICAL CONSIST SUB-MEMBER, Only linked upwards
 - 00=FREE locomotive, no CONSIST indirection/linking
- ALLOWS “CONSISTS of CONSISTS”. Uplinked means that Slot SPD number is now SLOT adr of SPD/DIR and STATUS of consist. i.e. is an Indirect pointer. This Slot has same BUSY/ACTIVE bits as TOP of Consist. TOP is loco with SPD/DIR for whole consist. (top of list). BUSY/ACTIVE: bit encoding for SLOT activity

2 BITS for BUSY/ACTIVE

- **D5-SL_BUSY**
- **D4-SL_ACTIVE**
 - 11=IN_USE loco adr in SLOT -REFRESHED
 - 10=IDLE loco adr in SLOT, not refreshed
 - 01=COMMON loco adr IN SLOT, refreshed
 - 00=FREE SLOT, no valid DATA, not refreshed

3 BITS for Decoder TYPE encoding for this SLOT

- **D2-SL_SPDEX**
- **D1-SL_SPD14**
- **D0-SL_SPD28**
 - 010=14 step MODE
 - 001=28 step. Generate Trinary packets for this Mobile ADR
 - 000=28 step/ 3 BYTE PKT regular mode
 - 011=128 speed mode packets
 - 111=128 Step decoder, Allow Advanced DCC consisting
 - 100=28 Step decoder ,Allow Advanced DCC consisting

Slots

Nr.	Description
0	dispatch
1 - 119	active locos
120 - 127	reserved for System and Master control
123	Fast Clock
124	Programming Track
127	Command Station Options

Programmer track

The programmer track is accessed as Special slot #124 (\$7C, 0x7C). It is a full asynchronous shared system resource. To start Programmer task, write to slot 124. There will be an immediate LACK acknowledge that indicates what programming will be allowed. If a valid programming task is started, then at the final (asynchronous) programming completion, a Slot read <E7> from slot 124 will be sent. This is the final task status reply.

Parameters

Name	Description	Bits/Range
ACK1¹	response code	0=busy/aborted, 1=accepted(OPC_SL_RD_DATA), 0x40=accepted blind(OPC_SL_RD_DATA), 0x7F=not implemented
PCMD	programmer command (0 will abort current operation)	0,WR,BM,TY1,TY0,OPS,0,0
	WR, Write/Read	1=Write, 0=Read
	MD, Byte mode	1=byte operation, 0=bit operation (if possible)
	OPS, Ops Mode	1=Mainline, 0=PT
PSTAT²	programmer error flags	0,0,0,0,D3,D2,D1,D0
	D3	1=User aborted
	D2	1=No read ack
	D1	1=No write ack
	D0	1=Programming track empty
HOPSA	operations mode programming	7 bit high addres, 0 if service mode (POM)
LOPSA	operations mode programming	7 bit low addres, 0 if service mode (POM)
CVH	CV#	0,0,CV9,CV8,0,0D7,CV7
CVL	CV#	0,CV6,CV5,CV4,CV3,CV2,CV1,CV0
DATA7	data	0,D6,D5,D4,D3,D2,D1,D0

¹) Note that the <7F> code will occur in Operations Mode Read requests if the System is not configured for and has no Advanced Acknowledgement detection installed.. Operations Mode requests can be made and executed whilst a current Service Mode programming task is keeping the Programming track BUSY. If a Programming request is rejected, delay and resend the complete request later. Some readback operations can keep the Programming track busy for up to a minute. Multiple devices, throttles/PC's etc, can share and sequentially use the Programming track as long as they correctly interpret the response messages . Any Slot RD from the master will also contain the Programmer Busy status in bit 3 of the <TRK> byte.

²) This <E7> response is issued whenever a Programming task is completed. It echos most of the request information and returns the PSTAT status code to indicate how the task completed. If a READ was requested <DATA7> and <CVH> contain the returned data, if the PSTAT indicates a successful readback (typically =0). Note that if a Paged Read fails to detect a successful Page write acknowledge when first setting the Page register, the read will be aborted, showing no Write acknowledge flag D1=1

Type codes

Byte Mode	Ops Mode	TY1	TY0	Meaning
1	0	0	0	Paged mode byte Read/Write on Service Track
1	0	0	0	Paged mode byte Read/Write on Service Track
1	0	0	1	Direct mode byteRead/Write on Service Track
0	0	0	1	Direct mode bit Read/Write on Service Track
x	0	1	0	Physical Register byte Read/Write on Service Track

Byte Mode	Ops Mode	TY1	TY0	Meaning
x	0	1	1	Service Track- reserved function
1	1	0	0	Ops mode Byte program, no feedback
1	1	0	1	Ops mode Byte program, feedback
0	1	0	0	Ops mode Bit program, no feedback
0	1	0	1	Ops mode Bit program, feedback

Multi Sense

Power and transponding information is brought with the **OPC_MULTI_SENSE** opcode.

Power

Variable	Value
ARG1	?
ARG2	?
ARG3	?
ARG4	?

Transponding

Variable	Value
ARG1	type, present/absent, and high part board address
ARG2	low part board address and Zone A to H in the lower 4 bits
ARG3	high part of transponding mobile decoder address
ARG4	low part of transponding mobile decoder address

Functions 9-12

Digitrax		Uhlenbrock	
Variable	Value	Variable	Value
OPC	OPC_IMM_PACKET	OPC	OPC_UHLI_FUN
REPS	0x24 for short and 0x34 for long address	Arg1	0x20
DHI	0x02 or 0x04 + high bits of address and functions	Arg2	Slot#

IM1	long address Lo	Arg3	0x07
IM2	long address Hi	Arg4	f9=0x10, f10=0x20, f11=0x40
IM3	0x20 + function bits & 0x7f		
IM4	0x00		
IM5	0x00		
IM1	short address		
IM2	0x20 + function bits & 0x7f		
IM3	0x00		
IM4	0x00		
IM5	0x00		

Functions 13-20

Digitrax		Uhlenbrock	
Variable	Value	Variable	Value
OPC	OPC_IMM_PACKET	OPC	OPC_UHLI_FUN
REPS	0x34 for short and 0x44 for long address	Arg1	0x20
DHI	0x02 or 0x04 + high bits of address and functions	Arg2	Slot#
IM1	long address Lo	Arg3	0x05=f12+f20+f28, 0x08=f13-f19
IM2	long address Hi	Arg4	f12=0x10, f20=0x20, f28=0x40, f13=0x01... f19=0x40
IM3	0x5E		
IM4	function bits & 0x7f		
IM5	0x00		
IM1	short address		
IM2	0x5E		
IM3	function bits & 0x7f		
IM4	0x00		
IM5	0x00		

Functions 21-28

Digitrax		Uhlenbrock	
Variable	Value	Variable	Value

OPC	OPC_IMM_PACKET	OPC	OPC_UHLI_FUN
REPS	0x34 for short and 0x44 for long address	Arg1	0x20
DHI	0x06 + high bits of address and functions	Arg2	Slot#
IM1	long address Lo	Arg3	0x09
IM2	long address Hi	Arg4	f21=0x01...f27=0x40
IM3	0x5F		
IM4	function bits & 0x7f		
IM5	0x00		
IM1	short address		
IM2	0x5F		
IM3	function bits & 0x7f		
IM4	0x00		
IM5	0x00		

Fast Clock

The system FAST clock and parameters are implemented in Slot#123 <7B>.

Use <EF> to write new clock information, Slot read of 0x7B,<BB><7B>.., will return current System clock information, and other throttles will update to this SYNC. Note that all attached display devices keep a current clock calculation based on this SYNC read value, i.e. devices MUST not continuously poll the clock SLOT to generate time, but use this merely to restore SYNC and follow current RATE etc. This clock slot is typically “pinged” or read SYNC'd every 70 to 100 seconds , by a single user, so all attached devices can synchronise any phase drifts. Upon seeing a SYNC read, all devices should reset their local sub-minute phase counter and invalidate the SYNC update ping generator.

Parameters

Name	Description	Bits/Range
CLK_RATE	Clock rate	0=Freeze clock, 1=normal 1:1 rate, 10=10:1 etc, max VALUE is 7F/128 to 1
FRAC_MINSH/L	FRAC mins hi/lo are a sub-minute counter , depending on the CLOCK generator. Not for ext. usage.	This counter is reset when valid <E6><7B> SYNC msg seen.
256-MINS_60	This is FAST clock MINUTES subtracted from 256.	Modulo 0-59
256-HRS_24	This is FAST clock HOURS subtracted from 256.	Modulo 0-23
DAYS	number of 24 Hr clock rolls, positive count	
CLK_CNTRL	Clock Control Byte	D6- 1=This is valid Clock information, 0=ignore this <E6><7B>, SYNC reply
ID1/2	This is device ID last setting the clock.	<00><00> shows no set has happened, <7F> <7x> are reserved for PC access

Stationary Broadcast Command

Note that a 3 byte DCC track packet configured as:

<sync> ,<1011-1111>,<1000-D c b a > <ecb>

is a DCC Broadcast Address to Stationary decoders. Broadcast LocoNet Switch adr is then

<SW2>=<0,0,a,D-1,1,1,1>, <SW1>=<0,1,1,1-1,0,c,b>

Stationary Interrogate Command

The DCC packet

<sync>,<1011-1111>,<1100-D c b a> <ecb>

is an Interrogation for all DS54's. This causes a 2 LocoNet <B1> messages encoding both Output state and Input state, for each sensor adr a/b/c encodes. Interrogate LocoNet Switch adr is

<SW2>=<0,0,a,1-0,1,1,1>, <SW1>= <0,1,1,1-1,0,c,b>

This is generated by DCS100 at power ON, and scans all 8 inputs of all DS54's.

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